

Carolina Cloud for AWS Users

Mapping AWS instance vernacular onto Carolina Cloud. Informal — ping us on anything fuzzy.

What's different

No instance ladder. On AWS the family letter (c/m/r) fixes your RAM:vCPU ratio and the size fixes the amounts in doublings. On CC you request **vCPU, RAM, and disk independently**, any combination up to the physical node.

So the translation is: **read the vCPU and RAM off your AWS instance and ask for those two numbers**. The family letter collapses into “how much RAM” — a c8a.24xlarge with 512 GB (which on AWS forces you up to r8a) is just 96 vCPU, 512 GB here. vCPU means the same thing on both sides: one hyperthread, ½ a physical core.

The one thing you *do* pick is a **tier**, not a specific node:

- **General Purpose** — a production Zen 5 **Turin** node (the EPYC 9655, or our larger 128-core Turin box). This is your c8a/m8a/r8a equivalent, and where you'll likely run.
- **High Performance** — lands on one of the two Zen 4 Threadripper PRO nodes below (higher clocks). You don't choose which within the tier — but if you need a specific one pinned, we can stand up a dedicated instance on it and hand it over.

Two things that hold across both tiers: instances are **always dedicated, never shared** — no noisy neighbors, no vCPU oversubscription, so benchmarks are repeatable. And storage is **hot, persistent NVMe** attached to the box (not detached network block storage like EBS) — it survives restarts and you get local-disk latency/bandwidth.

The hardware

What your AWS benchmarks actually land on:

Tier	CPU	arch	Cores / Threads	RAM	L3	Mem channels	AVX-512 execution	AWS analog
General Purpose	EPYC 9655 (Turin)	Zen 5	96 / 192	512 GB	384 MB	8	True 512-bit (single cycle)	c8a/m8a/r8a
General Purpose	EPYC 9755 (Turin, ES)	Zen 5	128 / 256	512 GB	512 MB	4 (2 DPC)	True 512-bit (single cycle)	c8a/m8a/r8a
High Performance	Threadripper PRO 7975WX	Zen 4	32 / 64	128 GB	128 MB	2	Double-pumped 256-bit (two cycles)	c7a
High Performance	Threadripper PRO 7965WX	Zen 4	24 / 48	192 GB	128 MB	2	Double-pumped 256-bit (two cycles)	c7a

Picking a tier: coming off c8a / many-core throughput → **General Purpose** (production Zen 5 Turin, up to 256 threads). Clock- or latency-bound → **High Performance** (Zen 4 Threadrippers boost to ~5.3 GHz and can beat a c7a core).

Instance mapping

vCPU = threads, 1:1 with AWS. RAM is whatever you ask for — these just mirror the AWS defaults.

AWS instance	vCPU	RAM	Ask CC for	Tier
c8a.24xlarge	96	192 GB	96 vCPU, 192 GB	General Purpose
c8a.48xlarge	192	384 GB	192 vCPU, 384 GB	General Purpose
m8a.24xlarge	96	384 GB	96 vCPU, 384 GB	General Purpose
r8a.24xlarge	96	768 GB	96 vCPU, 768 GB	General Purpose
c7a.16xlarge	64	128 GB	64 vCPU, 128 GB	High Performance
c7a.12xlarge	48	96 GB	48 vCPU, 96 GB	High Performance

Cost

Pricing is à la carte — you pay for exactly the resources you request, summed. No SKU markup.

Resource	Unit	Price
vCPU (General Purpose or High Performance)	vCPU-hr	\$0.01
RAM	GB-hr	\$0.01
NVMe disk	GB-hr	\$0.0001
Object storage	TB-mo	\$9
Cold storage (optional backup)	TB-mo	\$1.55
Egress	—	Free

Because RAM is priced separately, going memory-heavy (the **r**-tier move on AWS) just adds \$0.01/GB-hr instead of bumping you to a pricier family. And egress is free — vs AWS's ~\$0.05–0.09/GB, which dominates data-heavy genomics bills.

Getting started

Grab free credits + a login, pick a tier (General Purpose if you're coming off **c8a**), and request **vCPU** / **RAM** / **disk**. Note that self-service only exposes the two CPU **tiers** to steer node choice — you don't pick an exact machine through the platform. If you want a specific box (say, pinned to the 9755 or a particular Threadripper), we can spin up a dedicated instance on it and hand it over.

We're also happy to help you benchmark and to set up custom hardware instances — NUMA node pinning, core/memory affinity, tuned topologies, whatever your pipeline needs, we'll try and configure it.